CNENGR202 Instrumentation and Control

Part 1 L6 – Feedback Control – Algorithms I

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Outline

- Introduction
- Open–Loop Control
- Closed–Loop Control
- Differential Feedback Control
- Transfer Function

Introduction - 1

Control Engineering is the science of changing the dynamical behaviour of physical systems in some desirable way

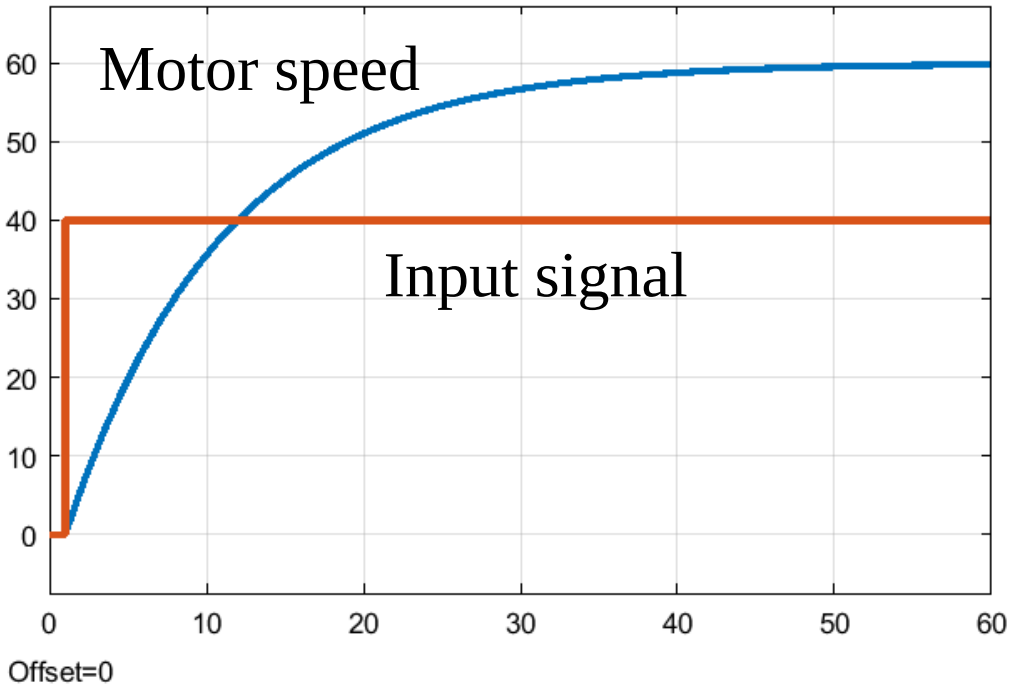
Introduction - 2

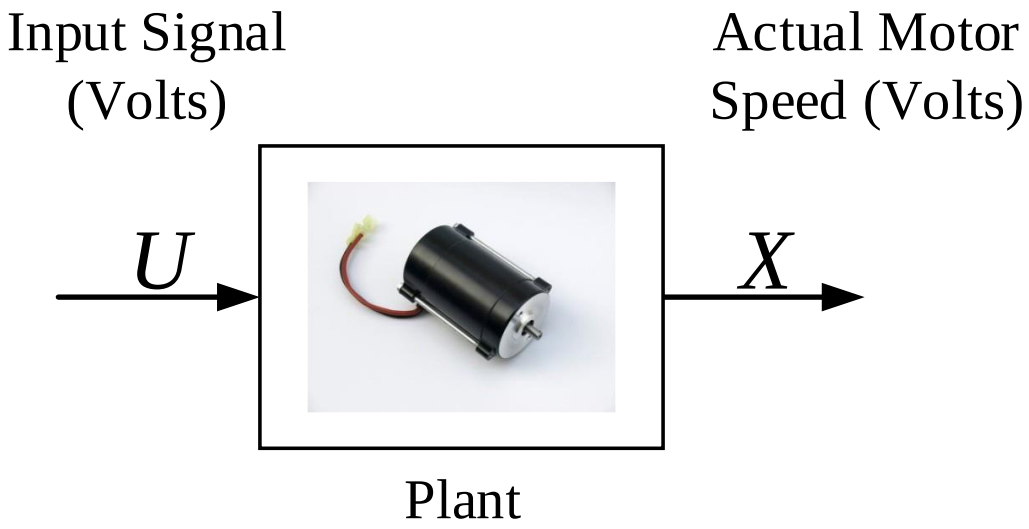
- Control engineering is a vital element of product and system design in all engineering disciplines
- Commercial aircraft have sophisticated autopilots that keep the aircraft on the correct heading and altitude
- Motor vehicles have active suspension systems, where computer controlled hydraulic jacks give improved handling and comfort
- Scale ranges from a single component (e.g. valve) to an entire system (e.g. power station, oil refinery, aircraft)
- The controller might be a mechanical device, a micro- processor or an algorithm programmed in software

Introduction - 3

- A common actuator is a DC motor which provides direct rotary motion or, via drums and cables, translational motion
- The non-turning stator has magnets, to establish a field across the turning rotor
- For the purposes of the lecturer, there is an input voltage U and a shaft speed output X

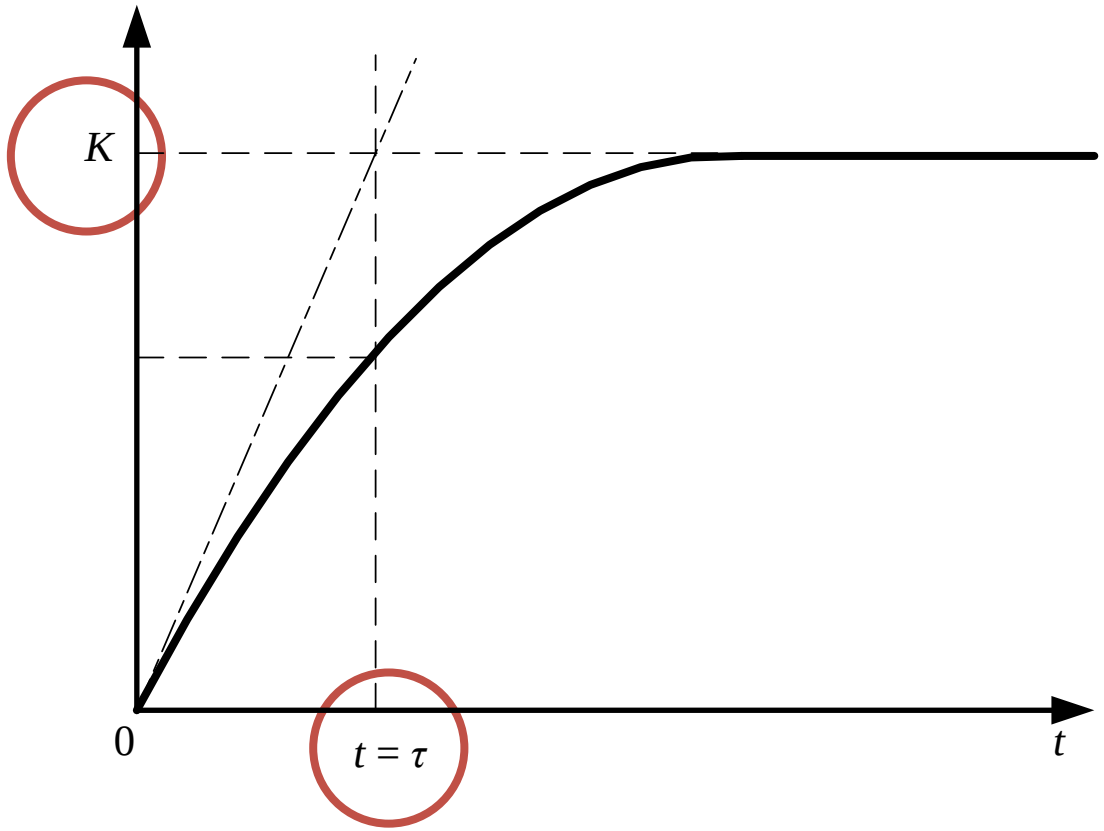
Introduction - 4





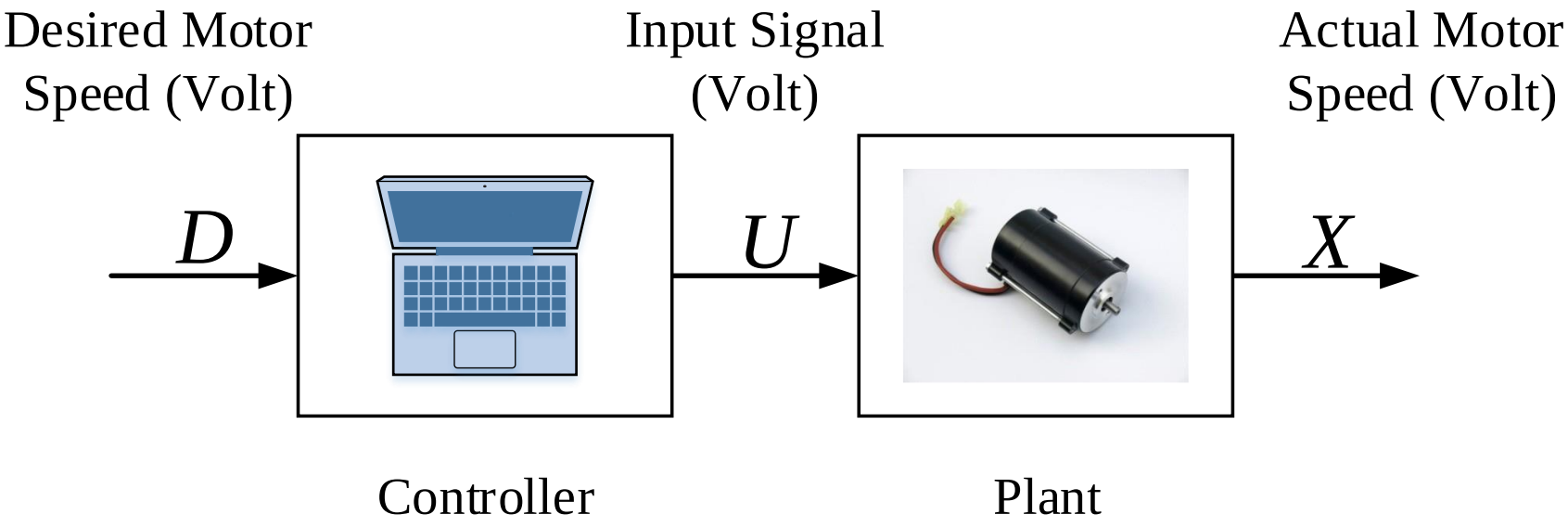
$$\tau \frac{dx}{dt} + x = Ku(t)$$

Introduction - 5

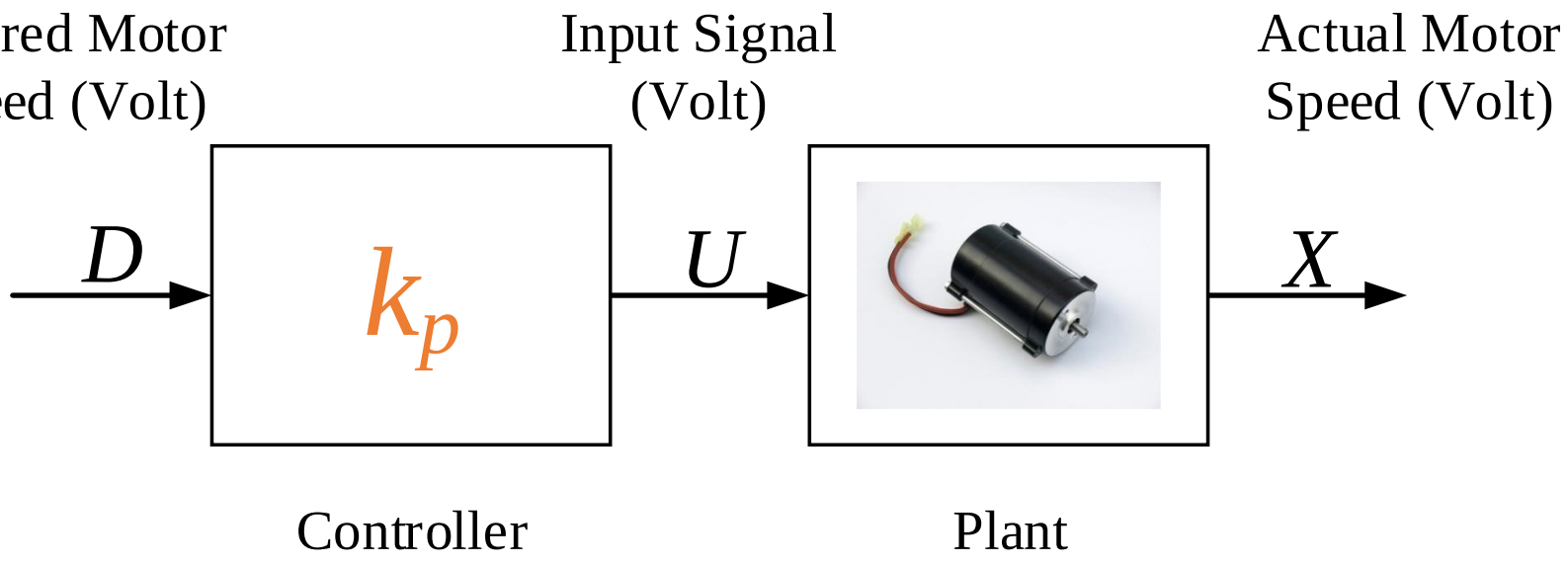


$$\tau \frac{dx}{dt} + x = Ku(t)$$

Open–Loop Control of a DC Motor - 1



Open–Loop Control of a DC Motor - 2



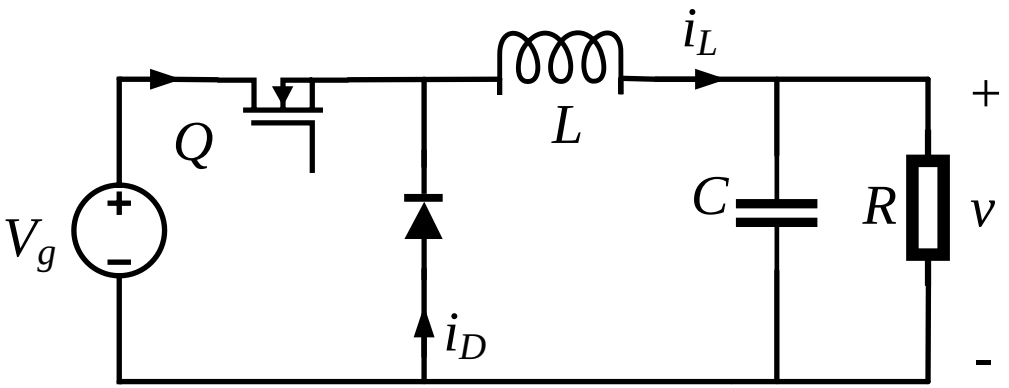
$$U = k_p \times D$$

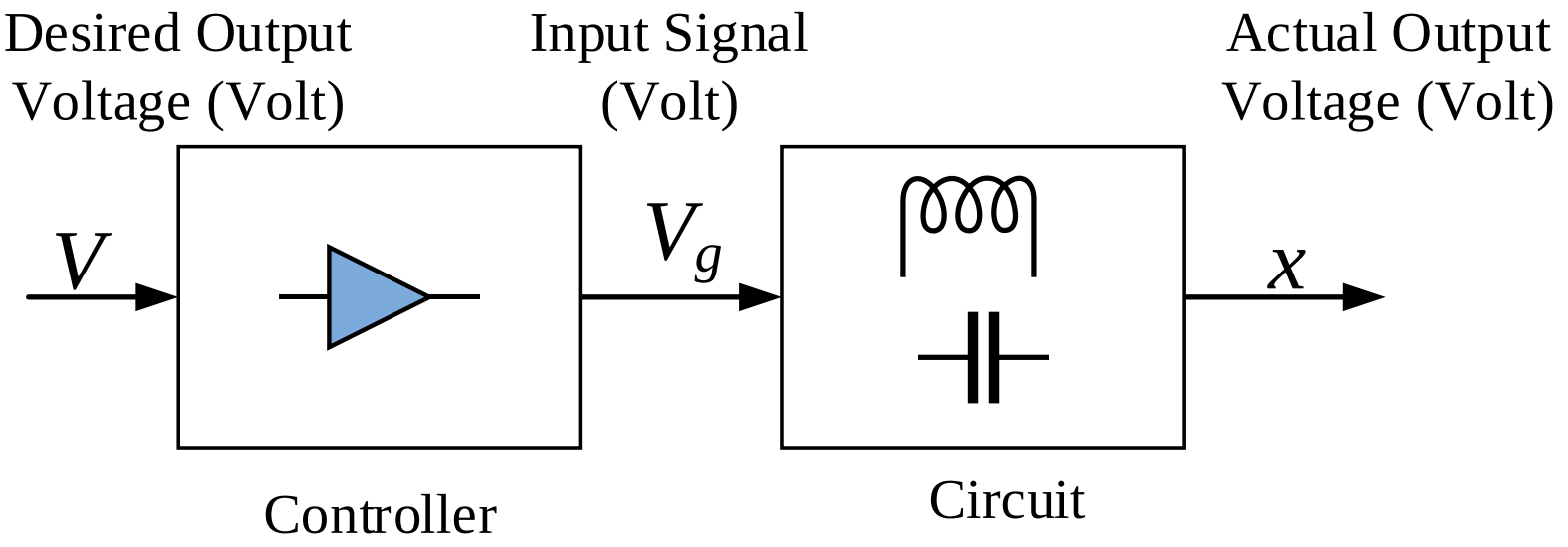
Control Algorithm

1. Select Algorithm (e.g. Proportional)
2. Optimise or “Tune” Algorithm

Example

- Open–Loop Control of a Buck converter



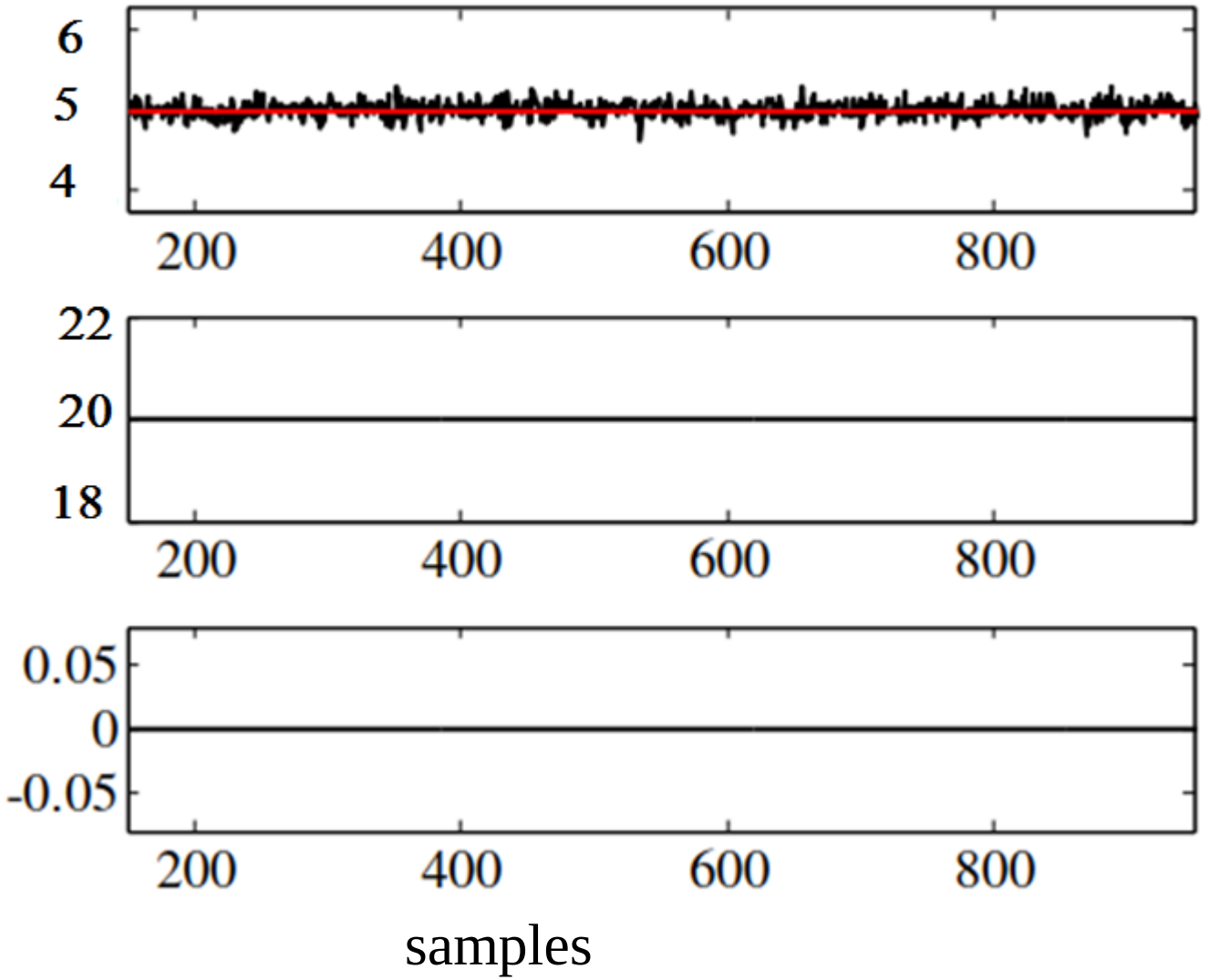


Open–Loop Control of a DC Motor - 3

Output

control Input

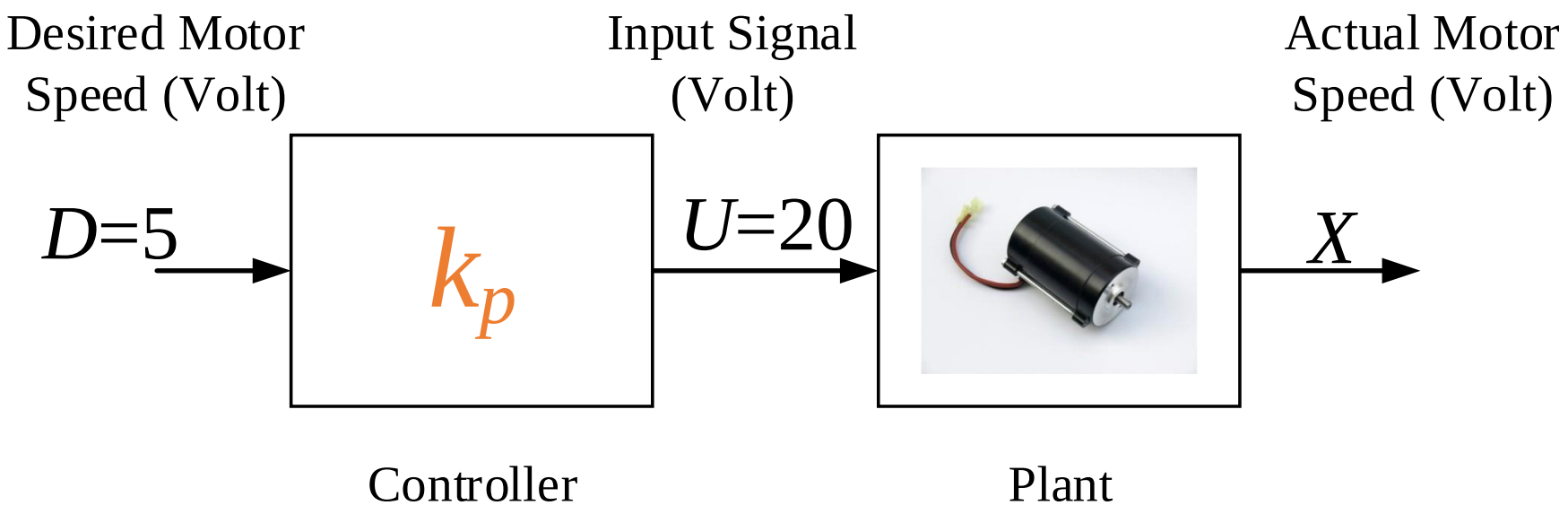
Disturbance (zero)



Example

Output	5V
Control Input	20V
Disturbance (zero)	0V

Open–Loop Control of a DC Motor - 4



$k_p = ?$

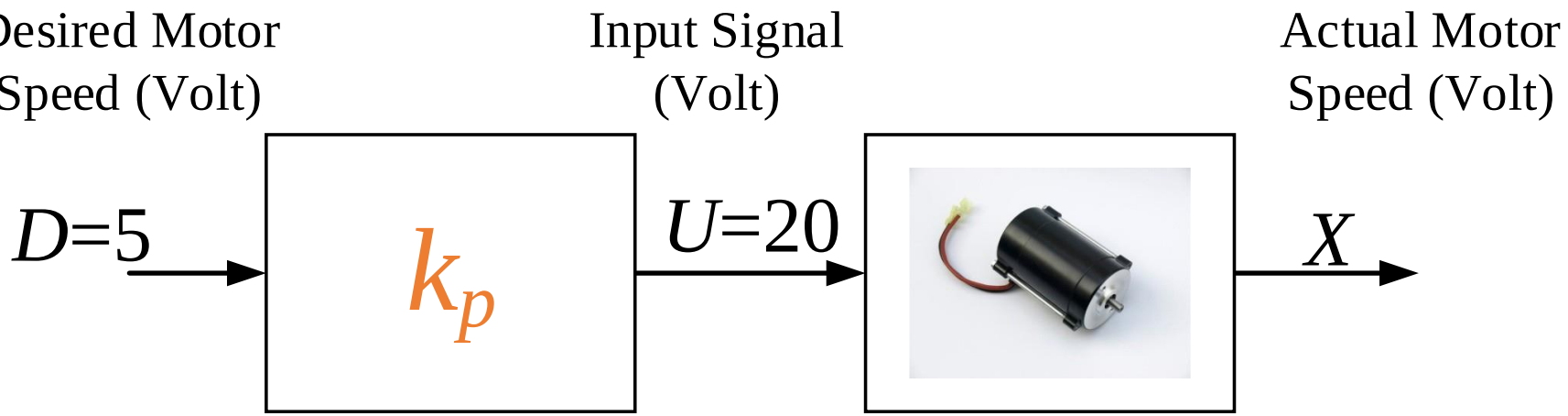
If U=20
D=5

"Steady state gain" of the DC motor

$K = ?$

"Control gain" or
"control coefficient" or
"control parameter"

Open–Loop Control of a DC Motor - 5



$$k_p = \frac{20}{5} = 4$$

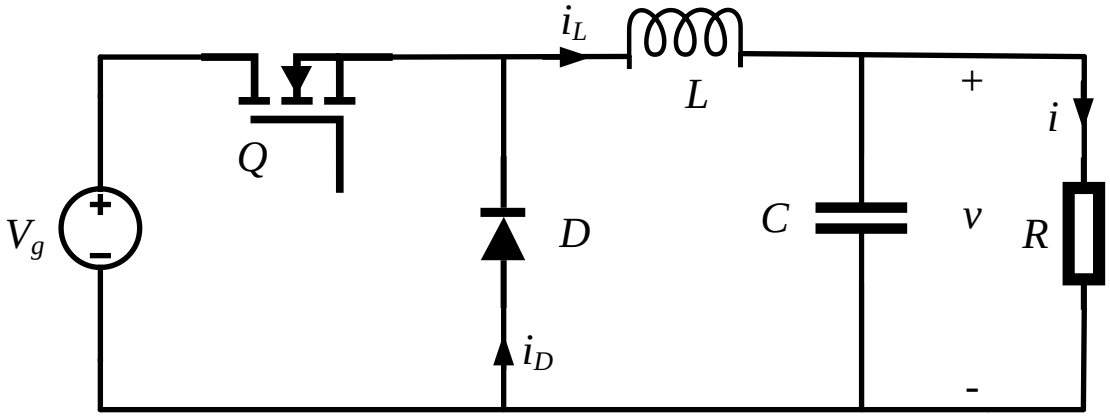
"Control gain" or
"control coefficient" or
"control parameter"

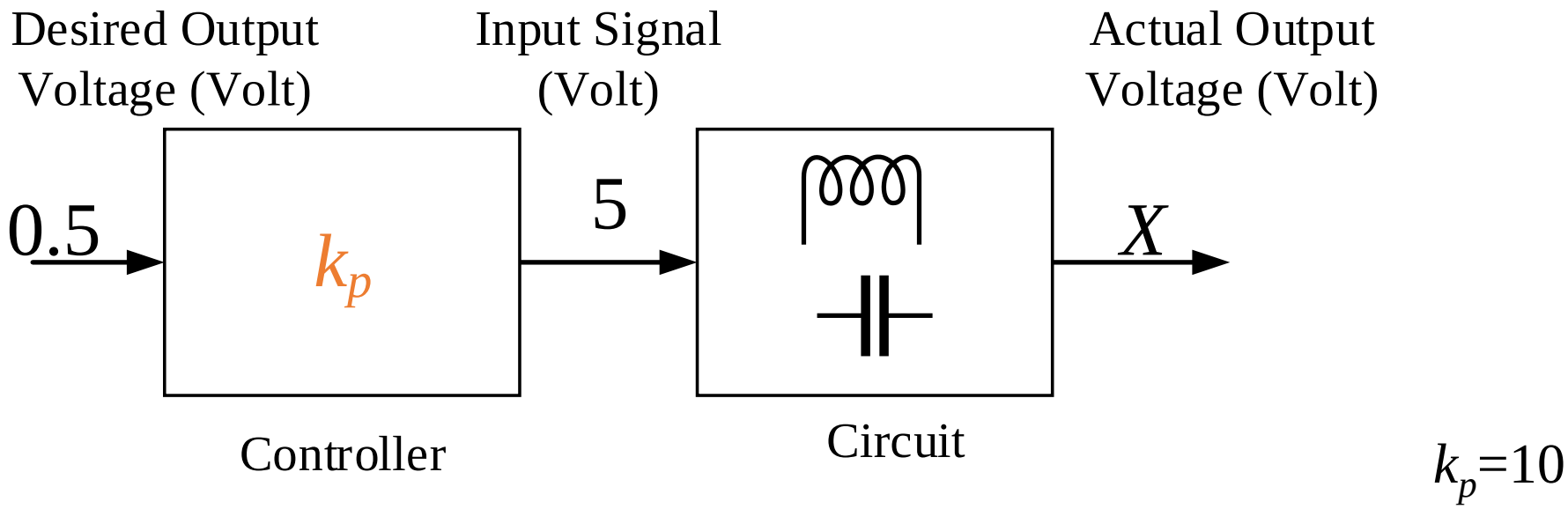
"Steady state gain"
of the DC motor

$$K = 0.25$$

Example

- Open–Loop Control of a Buck converter



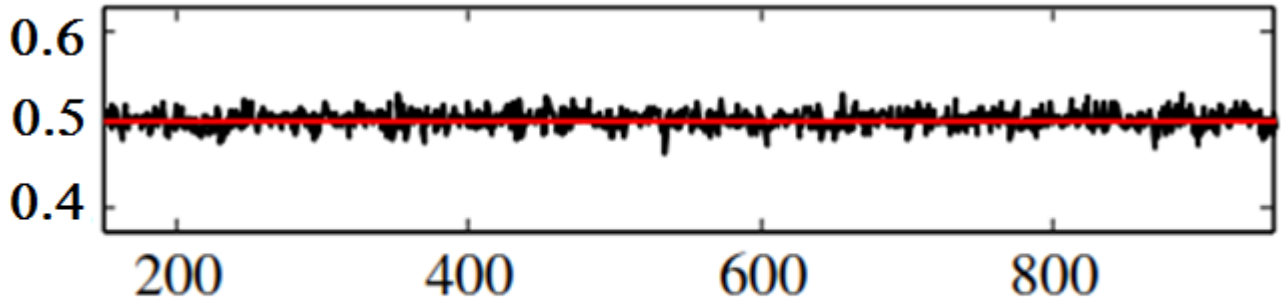


k_p=10

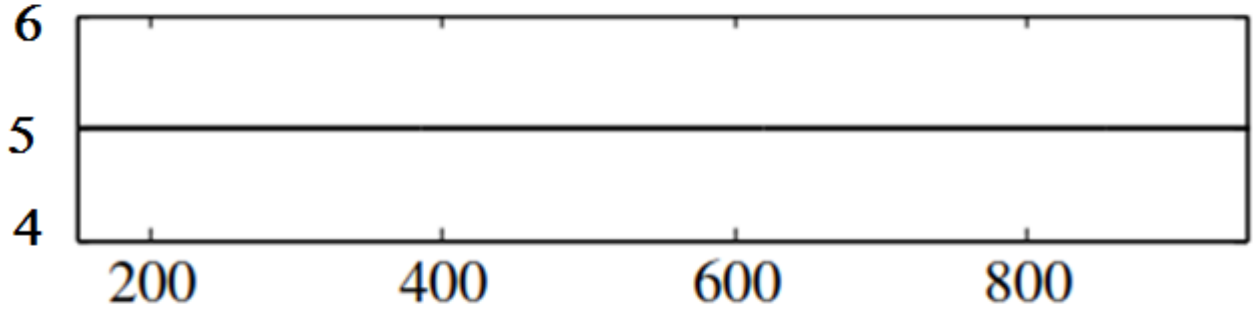
Open–Loop Control of a DC Motor - 6

"Steady state gain" of the DC motor is 0.1

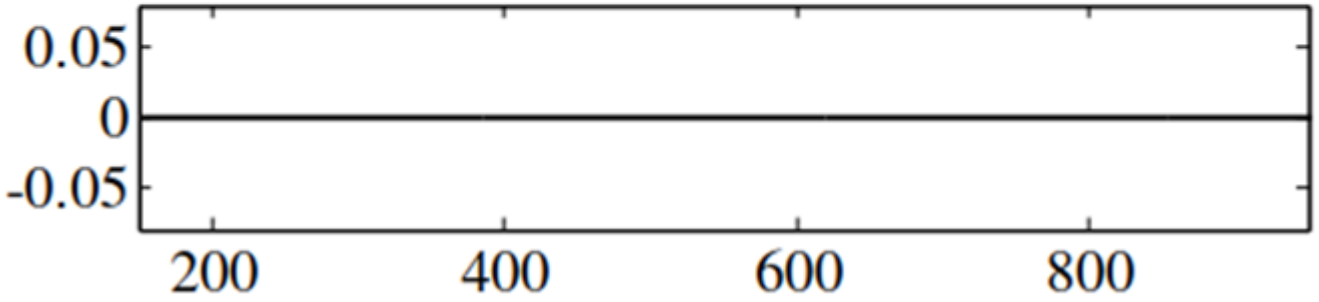
Output



Control Input



Disturbance (zero)



samples

Open–Loop Control of a DC Motor - 7

$$x(t \to \infty) = K \times u$$

$$\tau \frac{dx}{dt} + x = Ku(t)$$

Steady state Output voltage = 0.5 from the graph

Input signal $U = 5$ from graph

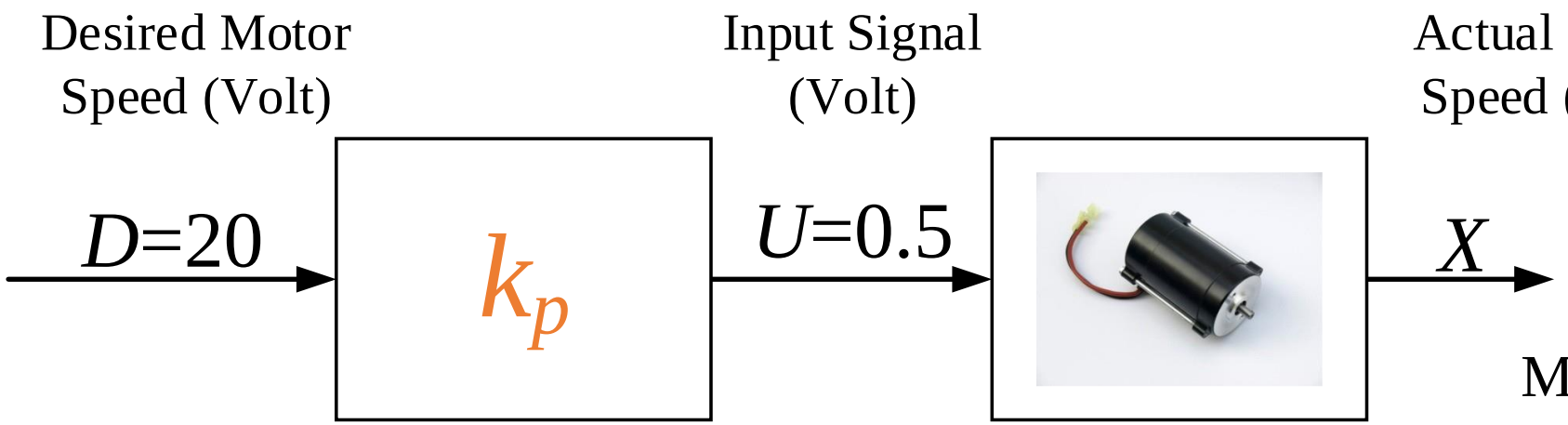
$$0.5 = K \times 5 \longrightarrow K = 0.1$$

"Steady state gain" of the Output voltage

For example, a particular output voltage might be say:

$$0.15 \frac{dx}{dt} + x = 0.1u(t)$$

Open–Loop Control of a DC Motor - 8



Motor Speed at **steady state** is 40*0.5=20 as required (assuming no disturbances)

$$k_p = \frac{0.5}{20} = 0.025$$

"Control gain" or
"control coefficient" or
"control parameter"

"Steady state gain" of the DC motor $K = 40$

Open–Loop Control of a DC Motor - 9

- User specifies a voltage representing the desired motor speed
- Controller could be an electronic amplifier that determines the input voltage
- For an open-loop proportional controller, the amplifier voltage is simply

$$U = k_p \times D$$

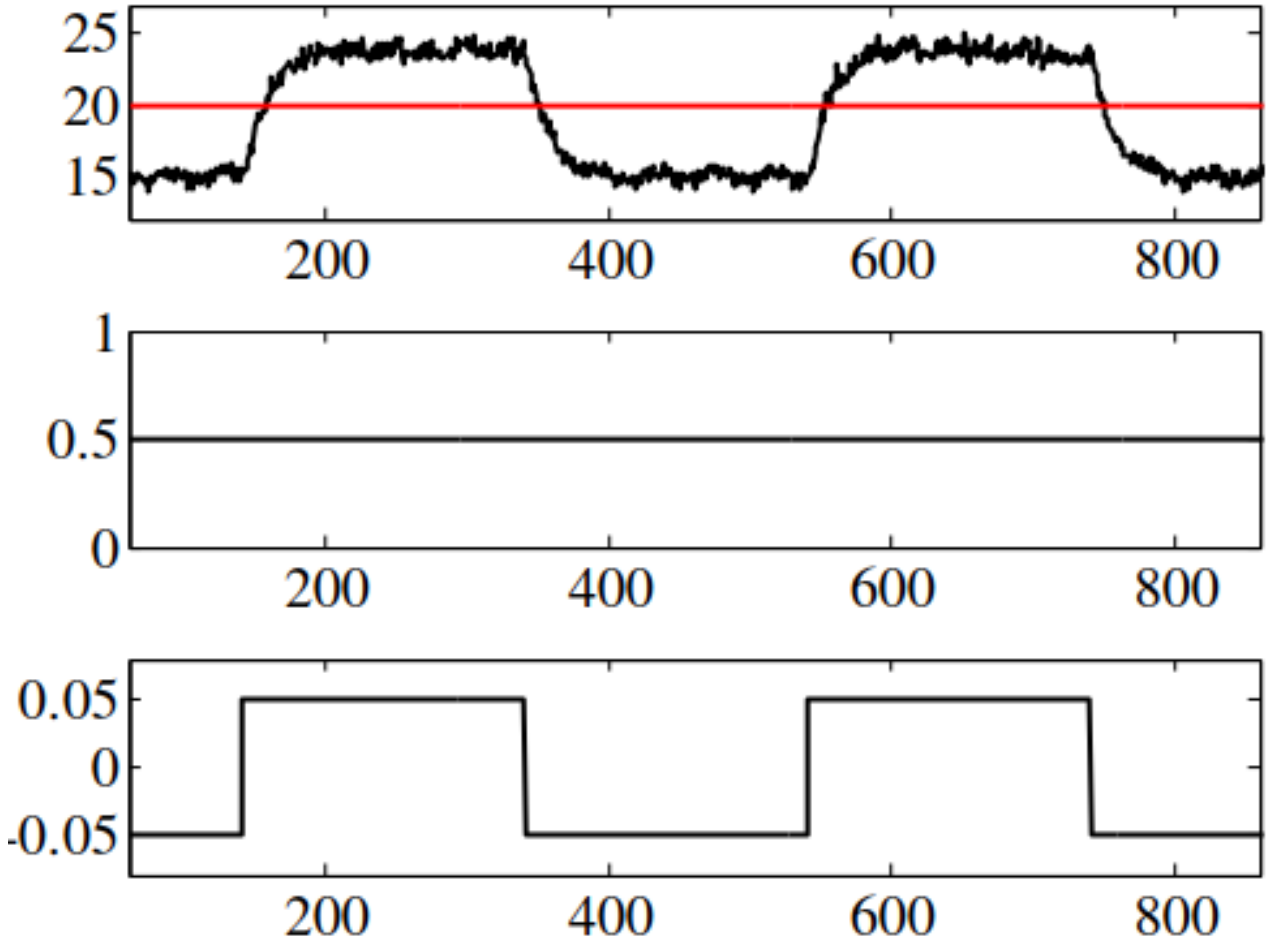
- However, the actual speed depends on the motor dynamics and any load disturbances

Open–Loop Control of a DC Motor - 10

Output

Control Input

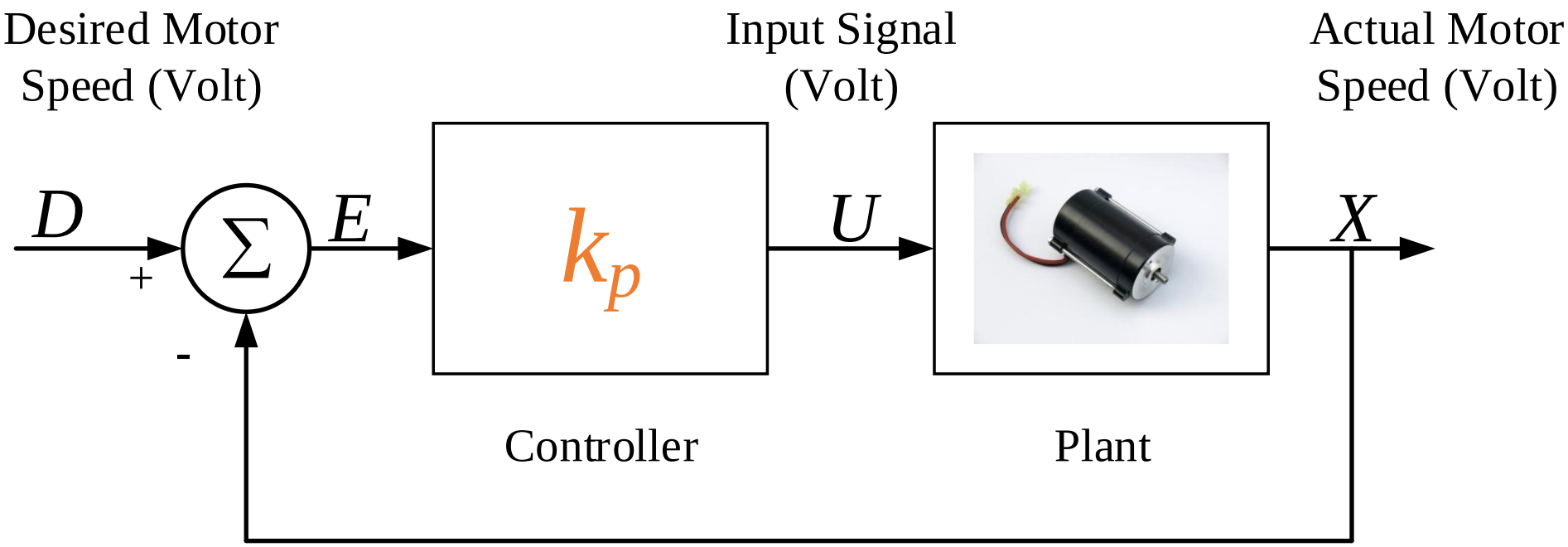
Disturbance



sample

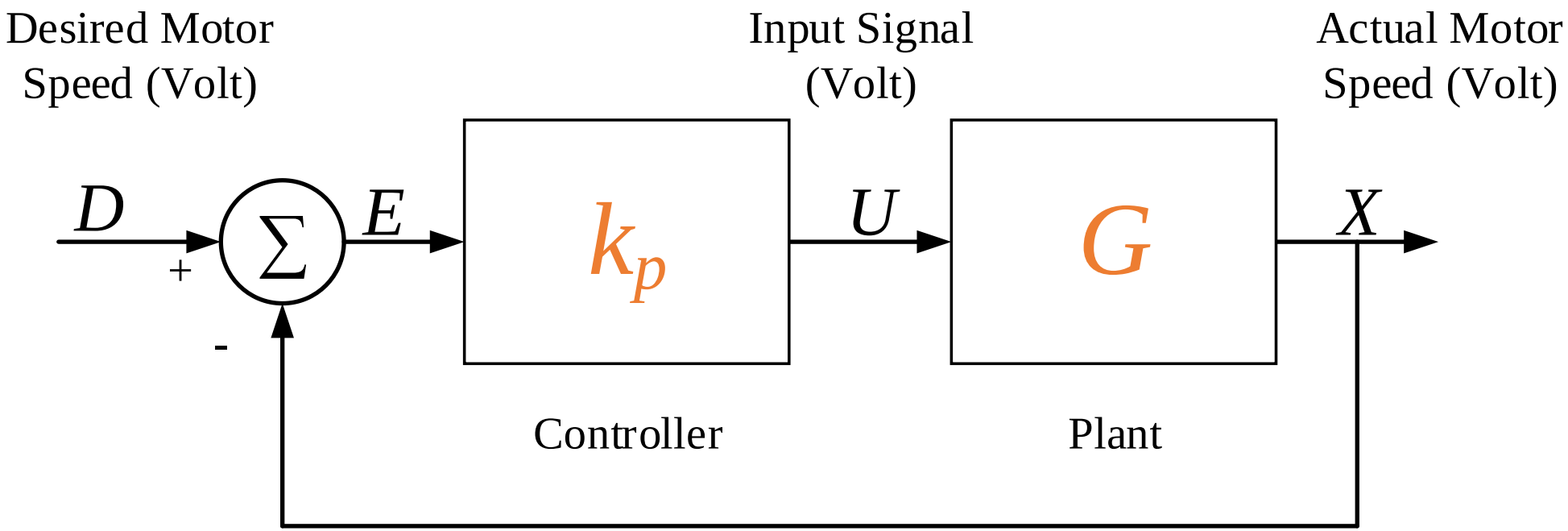
Need
A
New
Control
method

Closed–Loop Control of a DC Motor - 1



$$E = D - X \longrightarrow U = k_p E = k_p \left(D - X \right)$$

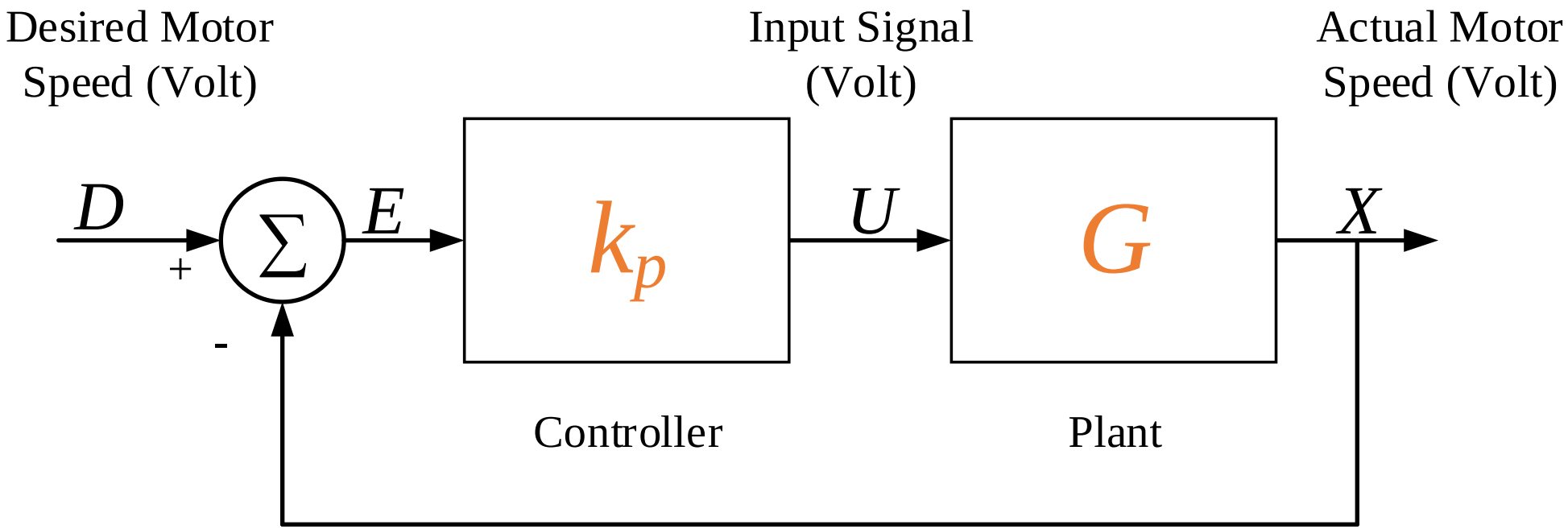
Closed–Loop Control of a DC Motor - 2



$$U = k_p\left(D - X\right) \qquad X = GU = Gk_p\left(D - X\right)$$

$$X = Gk_pD - Gk_pX \qquad X + Gk_pX = Gk_pD$$

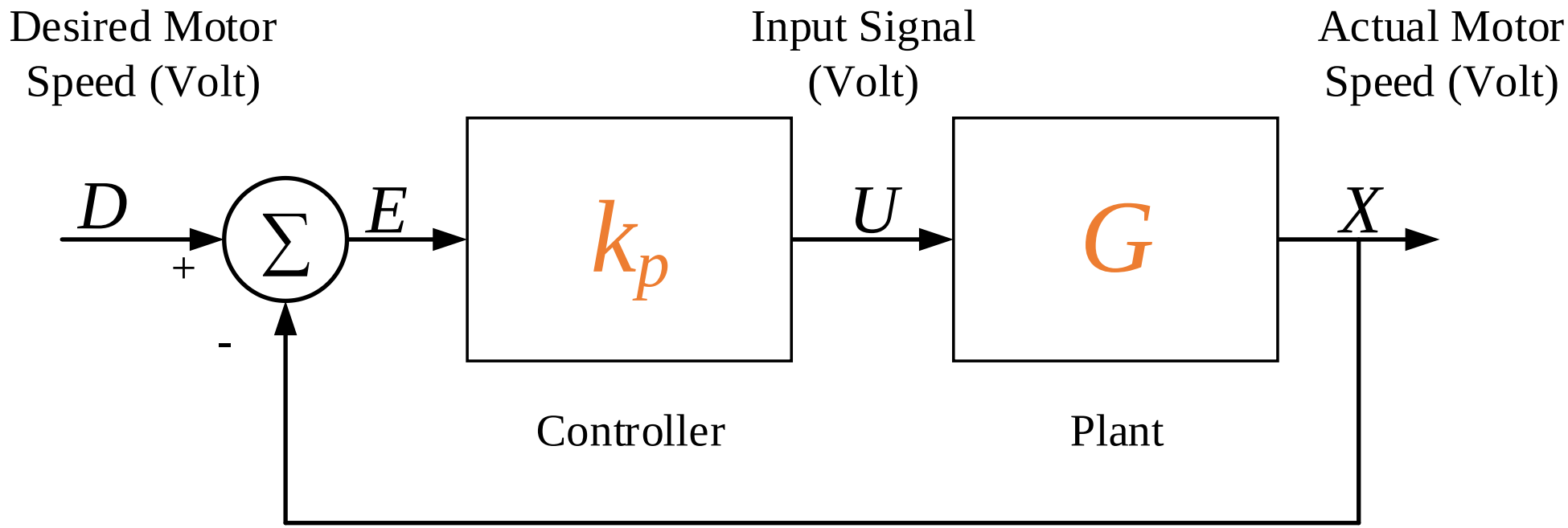
Closed–Loop Control of a DC Motor - 3



$$\left(1+Gk_p\right)X = Gk_pD$$

$$X = \frac{Gk_p}{1+Gk_p}D$$

Closed–Loop Control of a DC Motor - 4



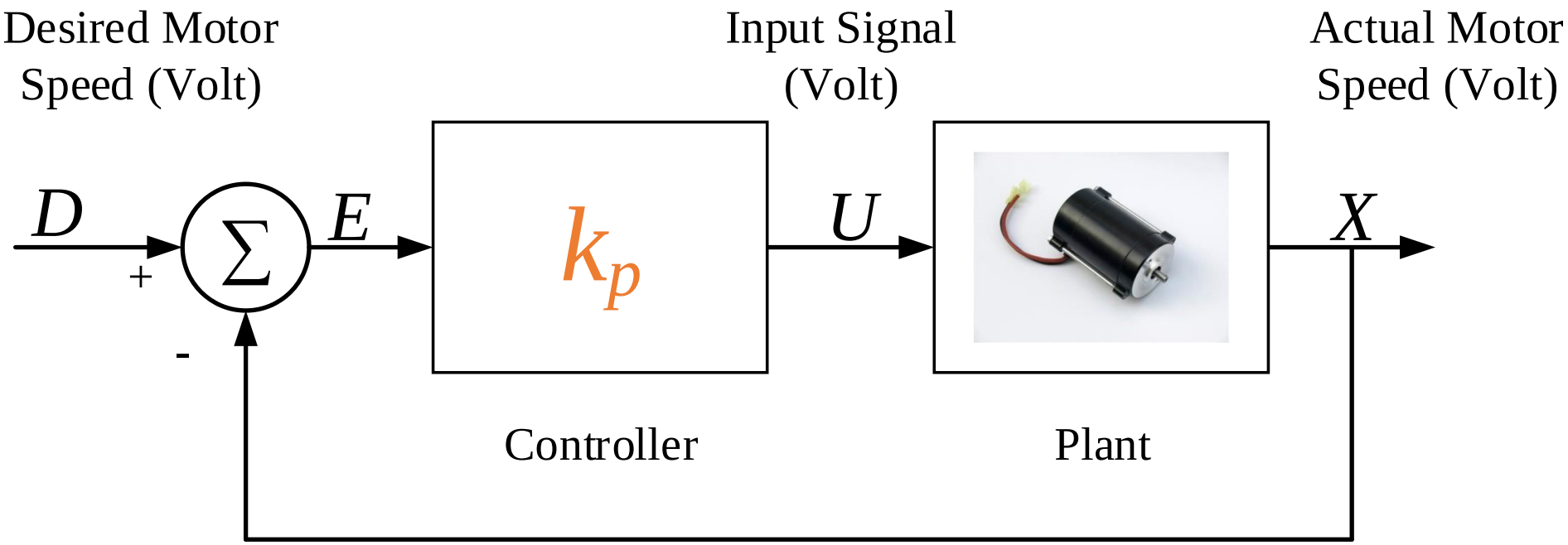
$$X = \frac{Gk_p}{1+Gk_p} D$$

For sufficiently high k_p then

$$X \approx D$$

"High gain control"

Closed–Loop Control of a DC Motor - 5



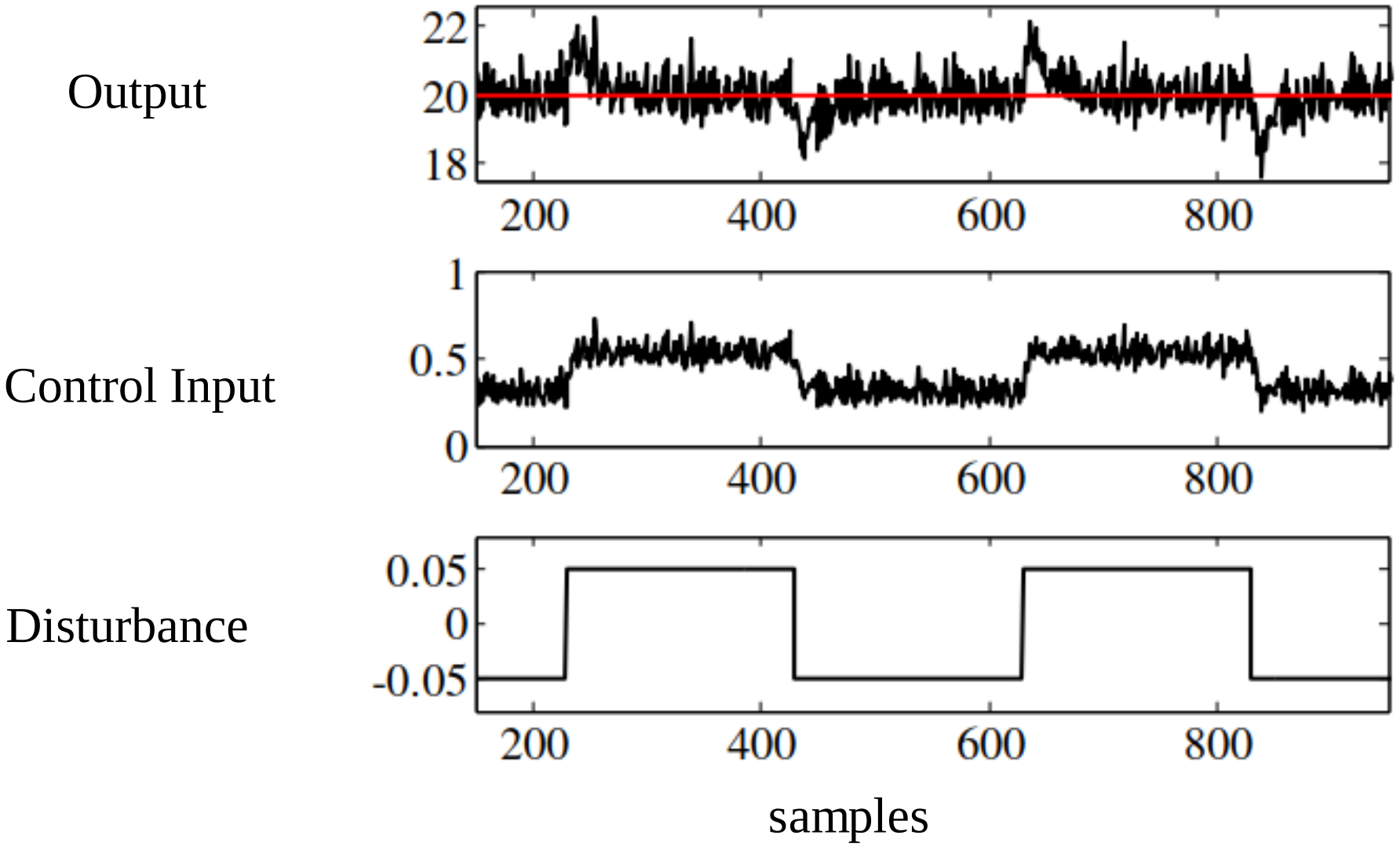
$$X = \frac{Gk_p}{1+Gk_p} D$$

For sufficiently high K_p then

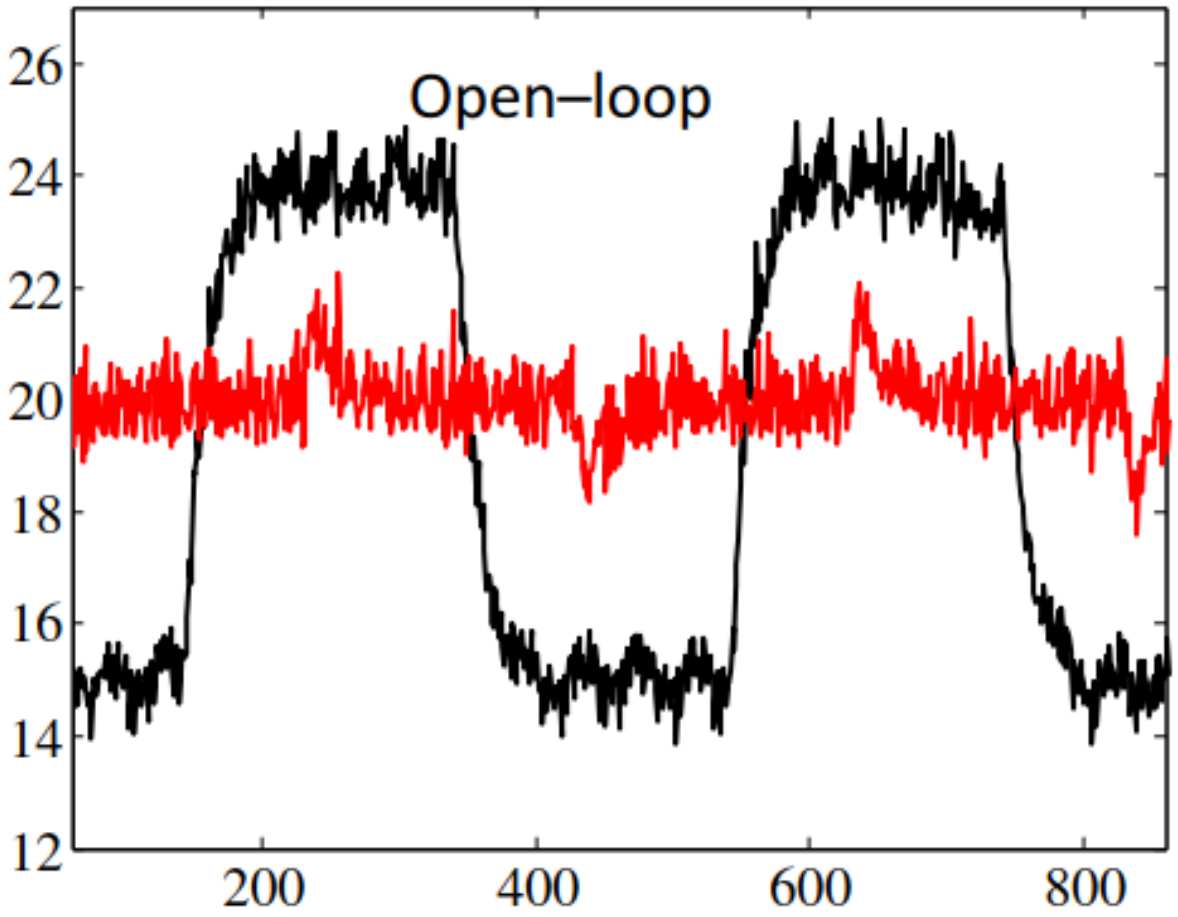
$$X \approx D$$

"High gain control"

Closed–Loop Control of a DC Motor - 6



Closed–Loop Control of a DC Motor - 7

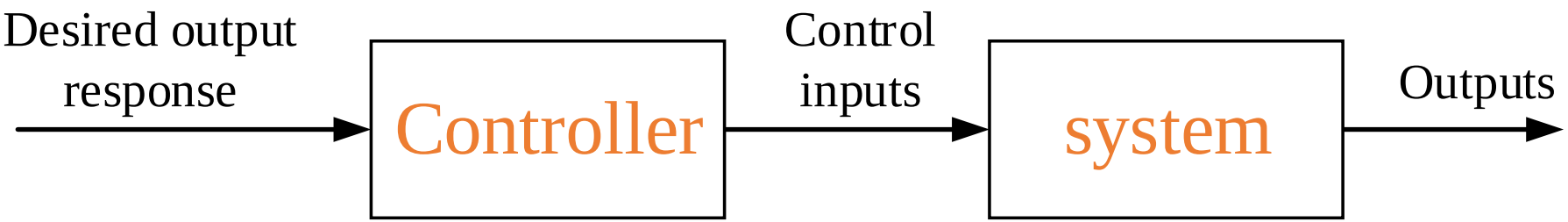


Closed–loop

Closed–Loop Control of a DC Motor - 8

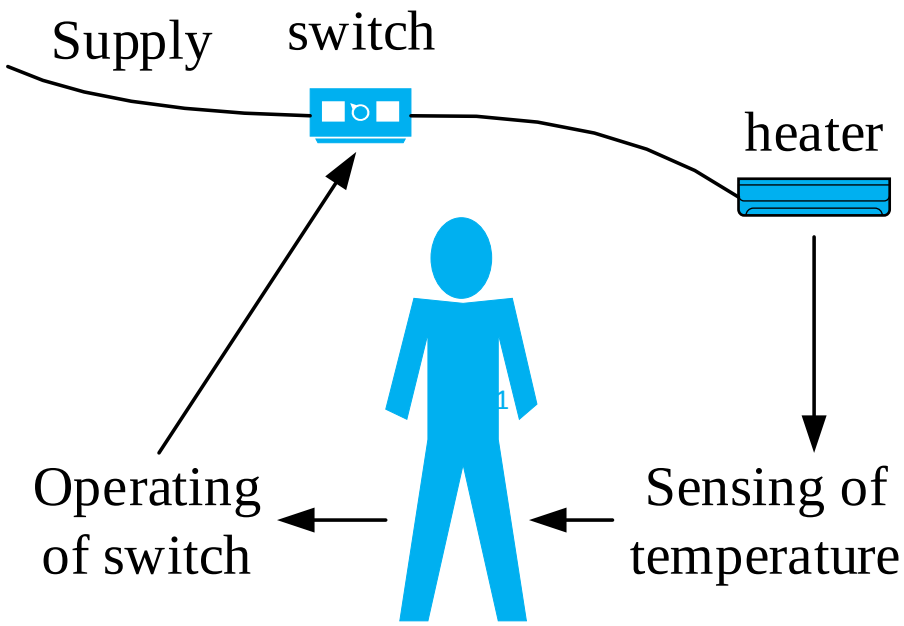
- Output is measured with a tachometer, usually a small permanent–magnet that returns a voltage proportional to shaft speed
- Closed–loop controller compares two voltages, i.e. representing the desired and actual speed
- An electronic amplifier produces a voltage proportional to this error
- Reduced sensitivity to load disturbances

Example: Open–Loop Control



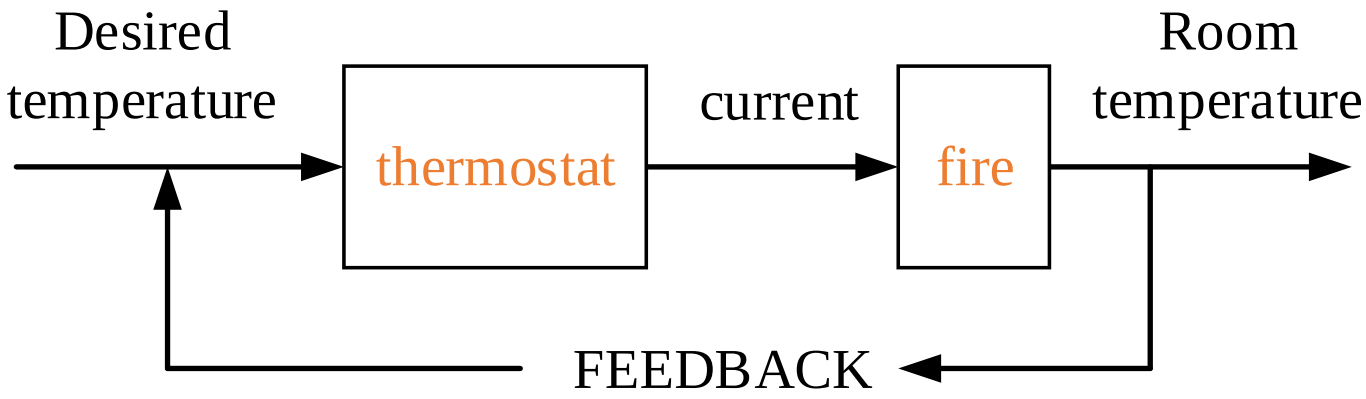
- Open–loop systems do **not** have feedback
- Example – an electric fire in a room
- Desired output response is a comfortable room temperature
- Controller is a human who turns heater on
- Without further human intervention, the room temperature (output) will eventually arrive at some constant level
- Room might still be too hot or too cold!

Example: Automatic feedback control

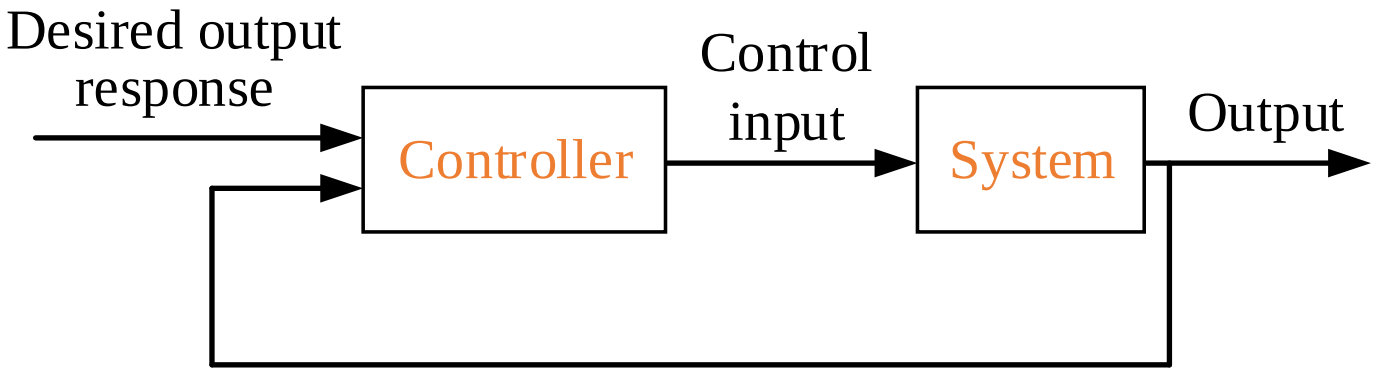


Feedback by human intervention

Automatic feedback control
using a thermostat



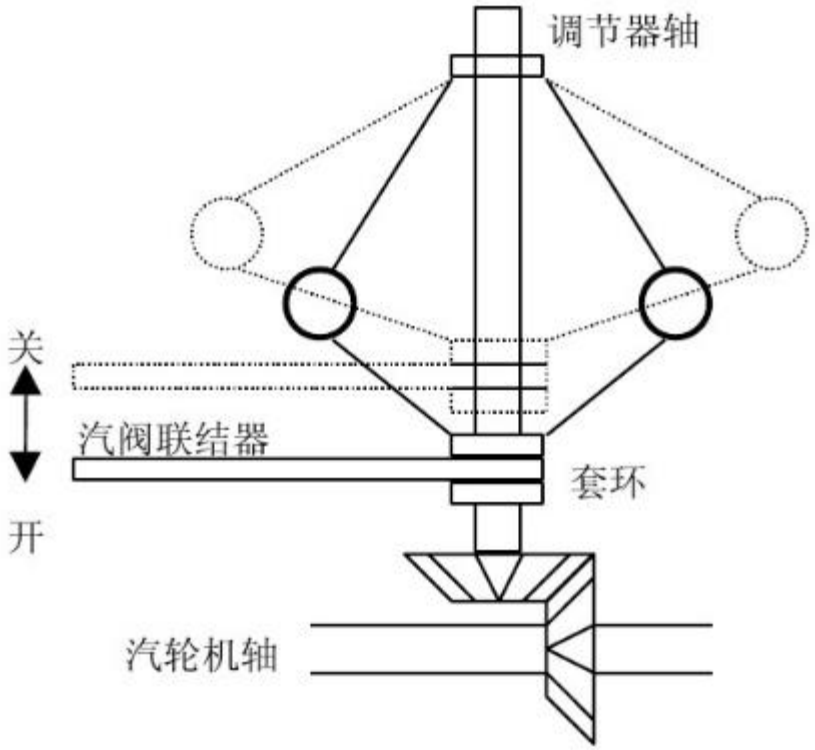
Example: Closed–Loop Control



- The system is also commonly called the plant or the process
- The desired output response is also called the set point, the command input or the reference level – essentially, this is how we want the system to behave
- The output is the (measured) response of the system – this is how the system actually behaves!
- The controller attempts to adjust the control input to minimise the difference between the actual output and the desired output

Watt's Flyball Governor

Invented in 1788 by James Watt to control his steam engine where it regulates the admission of steam into the cylinder(s)



James Clerk Maxwell's famous paper On Governors appeared in 1868

Advantage and disadvantage of Feedback

Advantages of Feedback

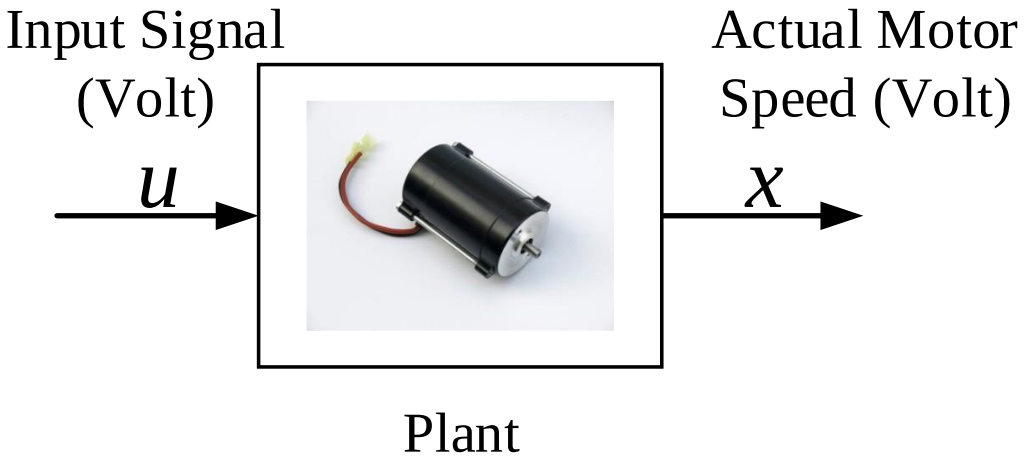
- Improved transient response
- Decreased sensitivity to modelling errors
- Decreased sensitivity to disturbances
- Ability to eliminate steady state errors
- Stabilisation of an unstable system

Disadvantages of Feedback

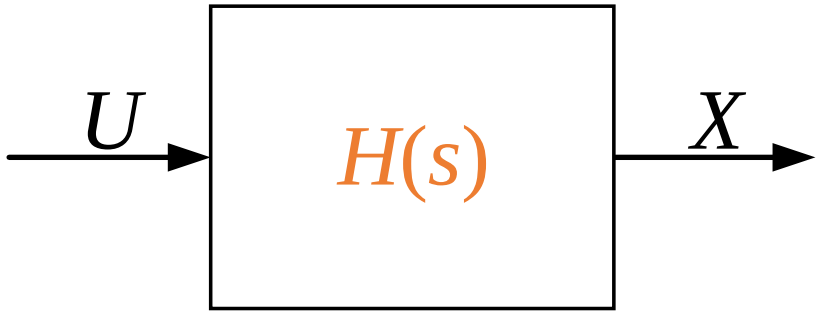
- Need to measure the output
- May be more difficult and expensive to design and construct
- Poorly designed feedback controllers can make things worse!

Example - 1

$$\tau \frac{dx}{dt} + x = Ku(t)$$



$$X = \frac{K}{\tau s + 1} U$$



If K=1 , τ=2 and u is an unit step function, then x=?

If K=1 , τ=0.5 and u = sin(4t), then x=?

Example - 2

If K=1 , τ=2 and u is a step function, then x=?

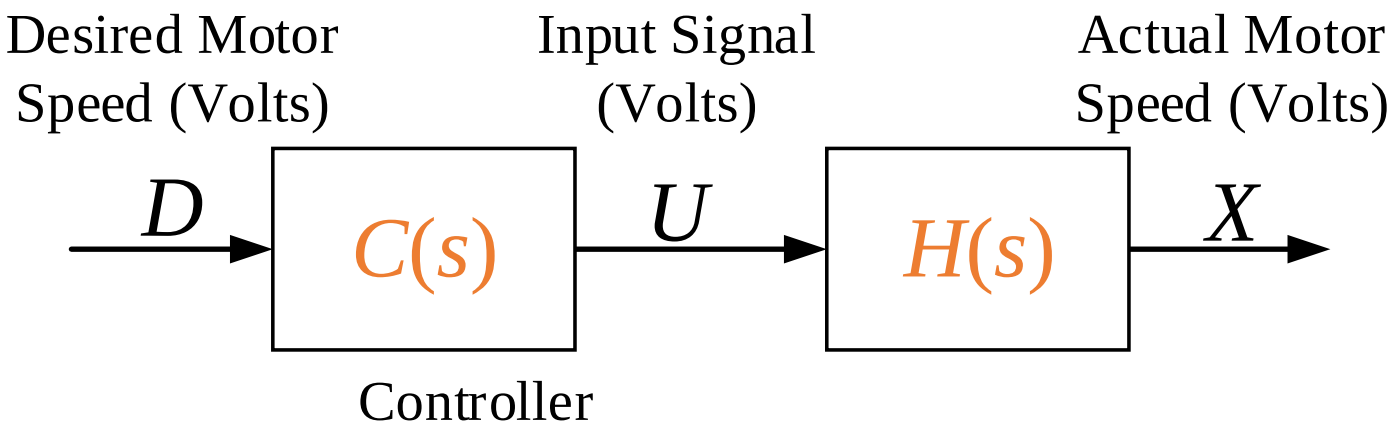
$$X = \frac{1}{2s+1}\frac{1}{s} = \frac{1}{s} - \frac{1}{s+0.5} \qquad x = 1 - e^{-0.5t}$$

If K=1 , τ=0.5 and u = sin(4t), then x=?

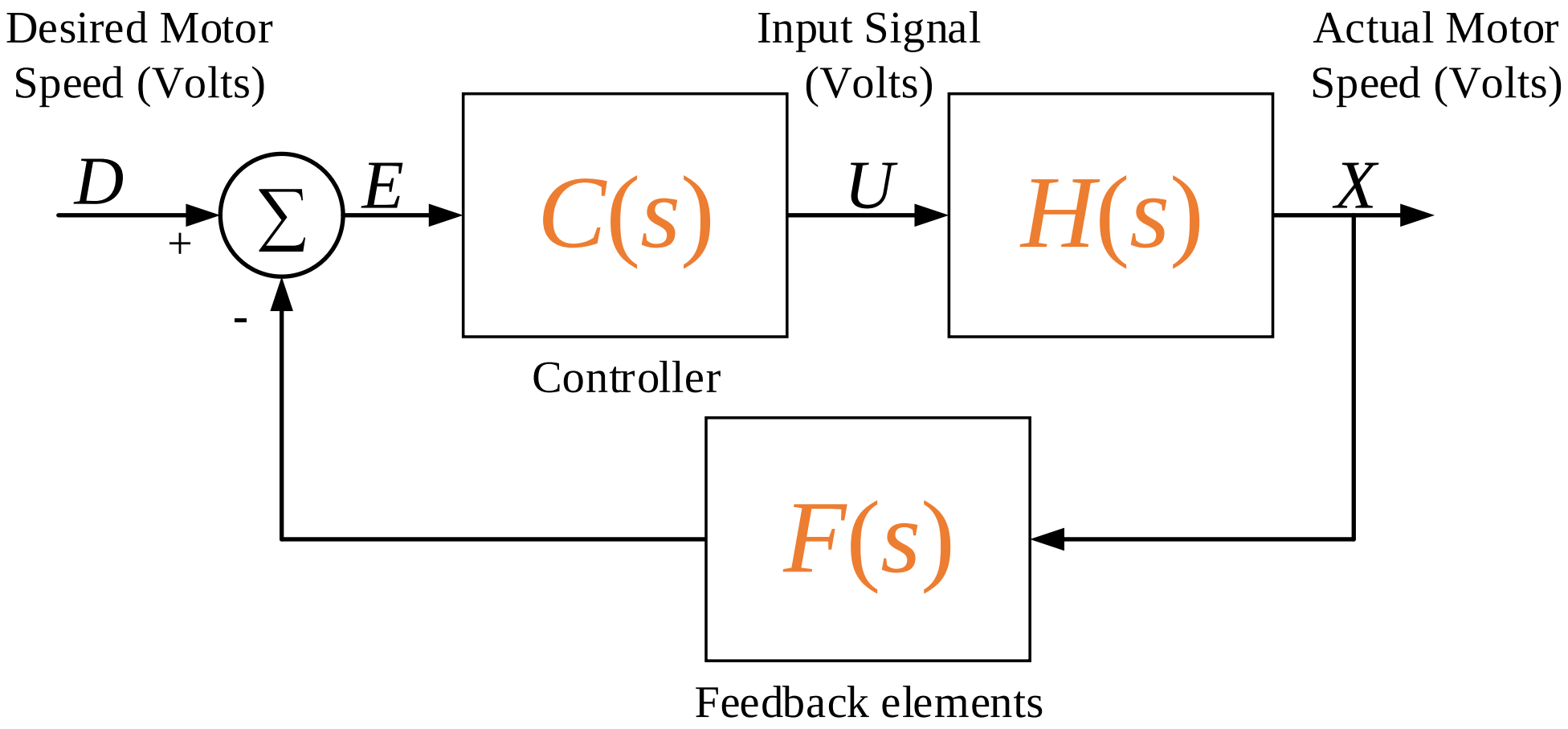
$$X = \frac{1}{0.5s+1}\frac{4}{s^2+4^2} = \frac{0.4}{s+2} - \frac{0.4\cdot s}{s^2+4^2} + \frac{0.2\cdot 4}{s^2+4^2}$$

$$x = 0.4e^{-2t} - 0.4\cos(4t) + 0.2\sin(4t)$$

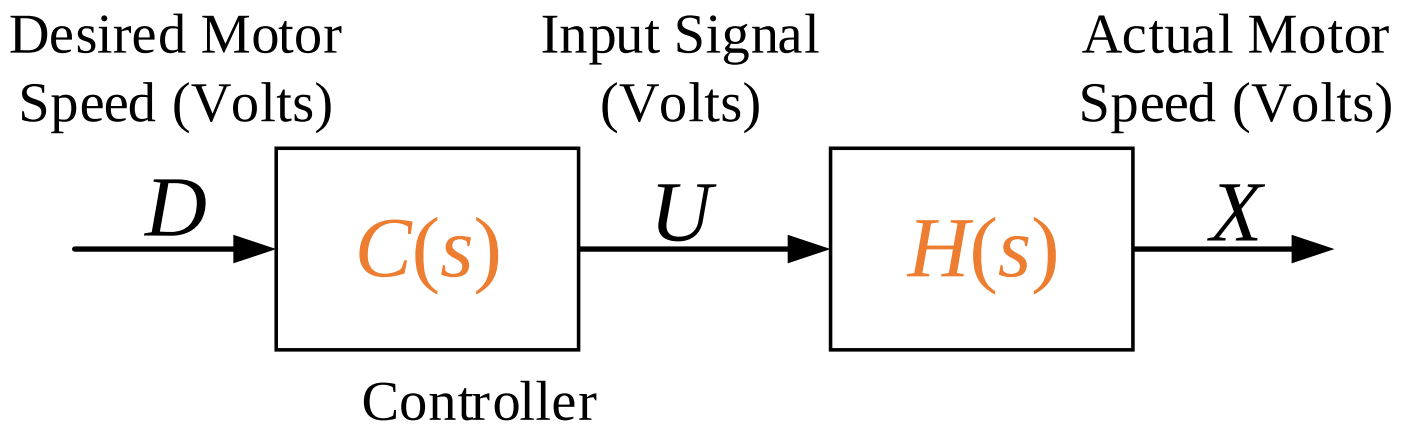
Differential Feedback - 1



Differential Feedback - 2

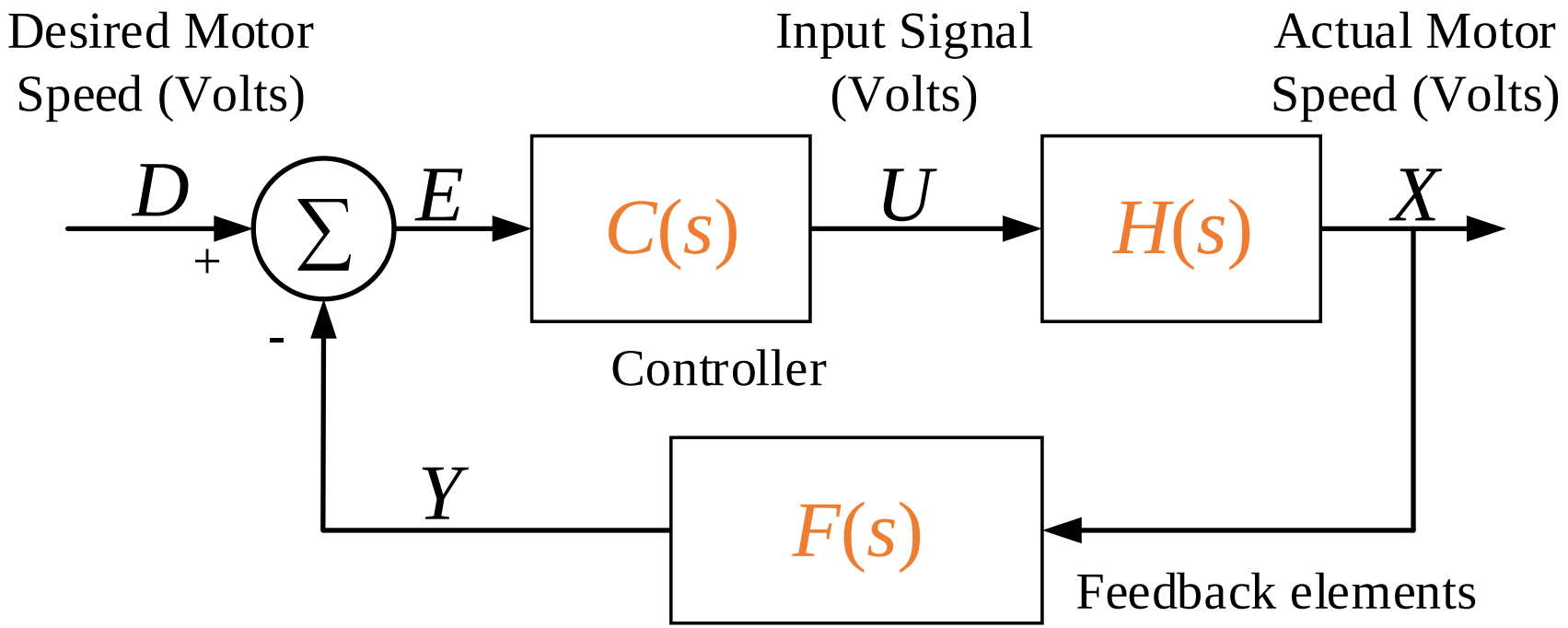


Open–Loop Transfer Function



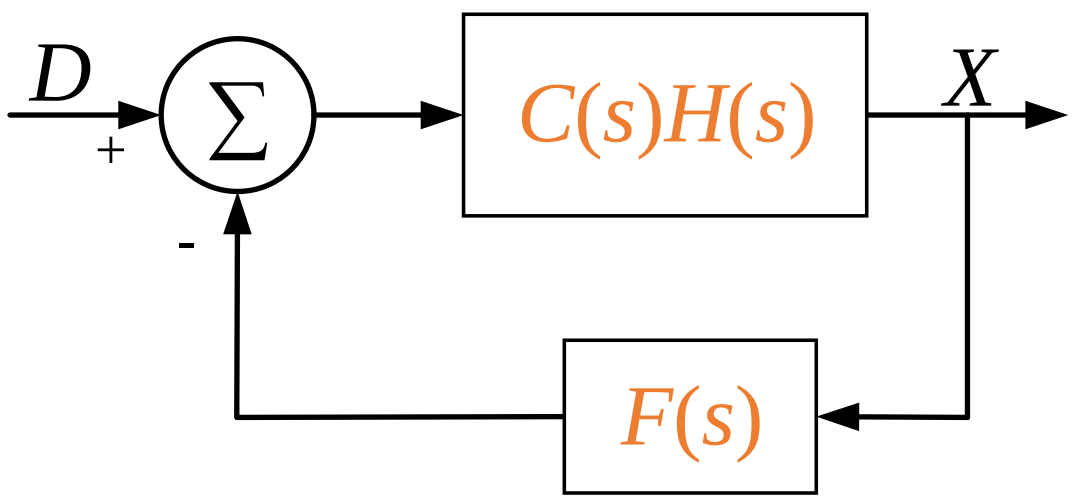
$$X = C(s)H(s)D$$

Closed–Loop Transfer Function



Either solve using algebra (as for the DC motor earlier) or use the Rules of Block Diagram Analysis

Using the Series and Negative Feedback rules



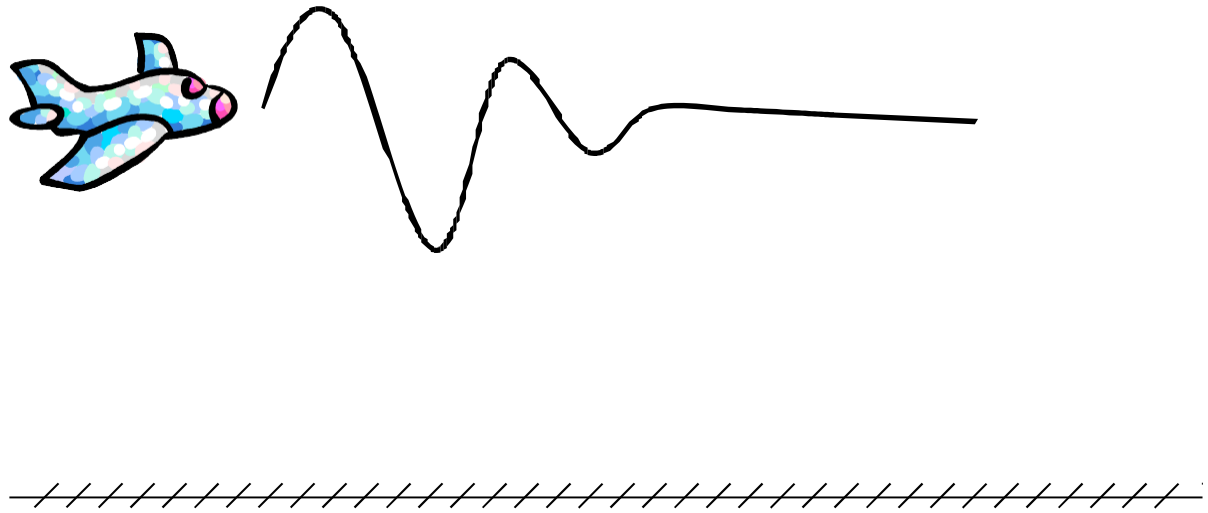
$$D \rightarrow \frac{C(s)H(s)}{1+\left(C(s)H(s)\right)\times F(s)} \rightarrow X$$

Control Objectives

- The closed–loop system should be **stable**
- The output should track the desired output response in the **steady state**, i.e. there should be no steady state error
- The transient behaviour of the system should be satisfactory
- The control system should reject **disturbances**

Question 1

- How do these objectives relate to the altitude of a passenger aircraft on final approach to the runway in fog?



Question 2

- How do the flight control objectives of a military aircraft differ from a commercial passenger jet?

Summary

- Closed Loop Transfer Function

$$X = \frac{C(s)H(s)}{1+C(s)H(s)F(s)}D$$

- We want unity steady state gain. To find the steady state gain set $s = 0$ in the Transfer Function above
- Characteristic Equation $1+C(s)H(s)F(s)=0$
- The roots (poles) of this equation define the transient response and stability
- The system or plant $H(s)$ might already have been built. In the rest of this module, we need to choose $C(s)$ and $F(s)$ to meet the control objectives.